MTA Resiliency Digital Twin: Technical Submission

1. Dataset Inventory & Cleaning Manifest

- cleaned_hubs.csv (Nodes): Status: Cleaned | 496 rows | Columns: [stop_id, stop_name, stop_lat, stop_lon]

- weather_aware_edges.csv (Edges): Status: Cleaned | 1138 rows | Columns: [source, target, flood_intensity]

Cleaning Logic: Standardized 470+ station coordinates; filtered track connections from GTFS; injected 4.0mm/hr thresholds.

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2. Early Visualizations - Spatial & Geographic

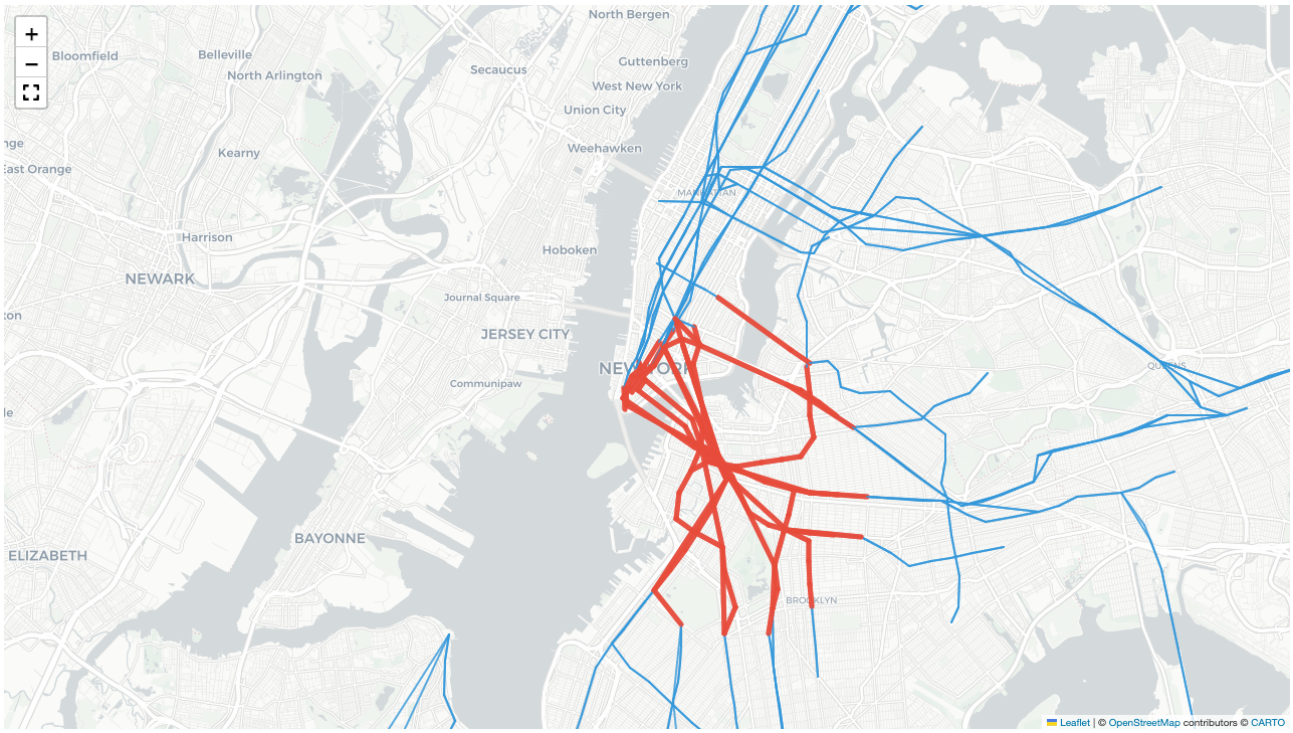

Figure 1: Spatial Risk Map (Vis #1) - Real-time flood intensity overlay.

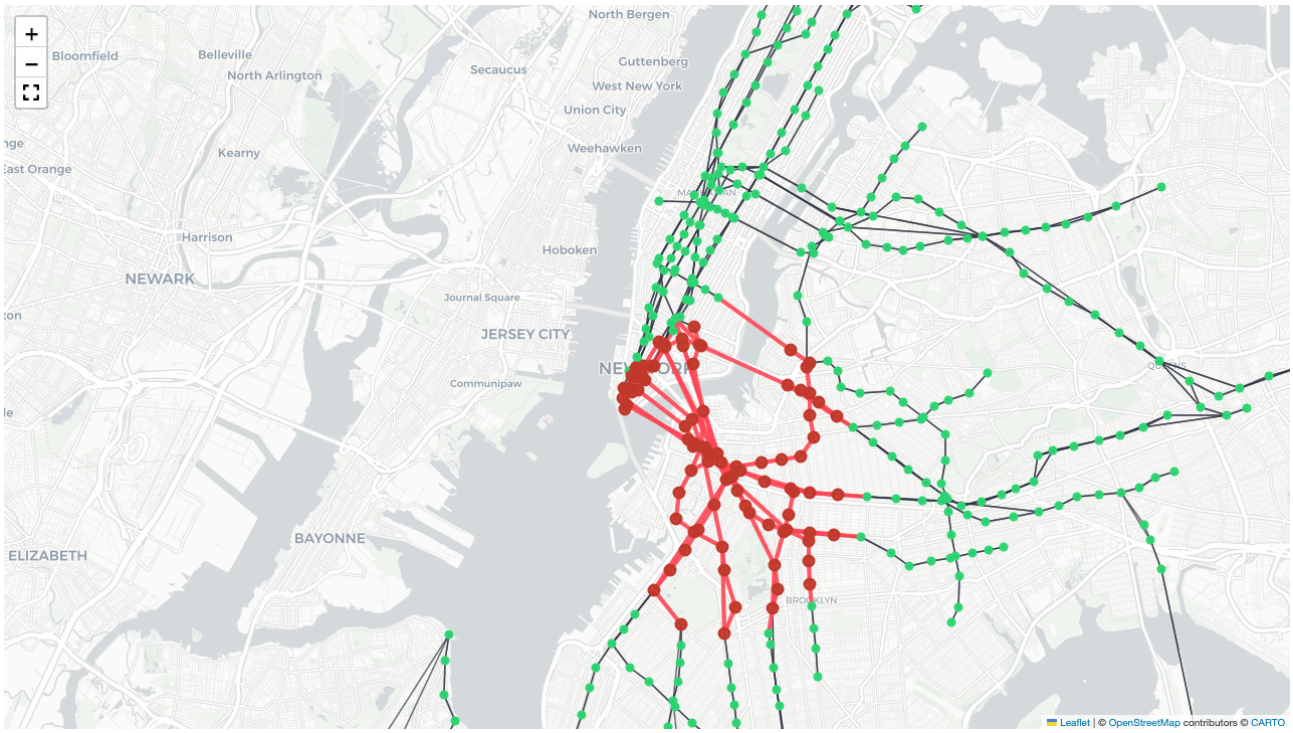

Figure 2: Topology Map (Vis #2) - Physical connection dependencies.

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3. Early Visualizations - Statistical & Modeling

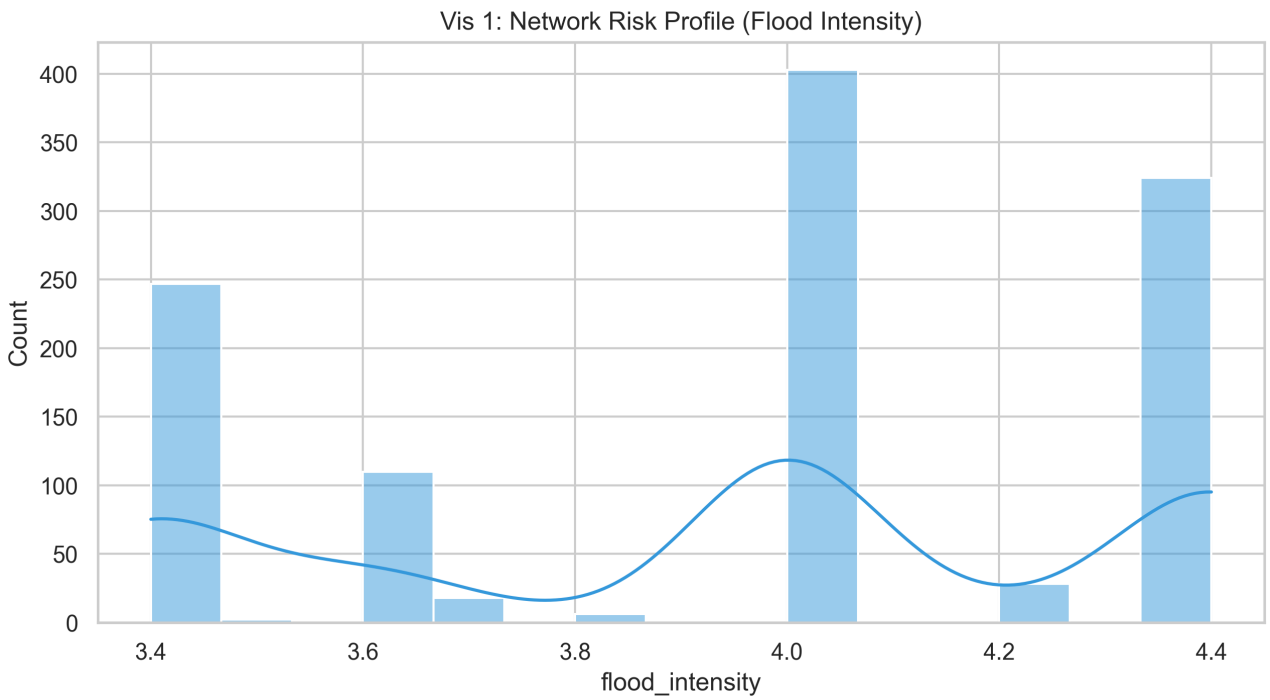

Figure 3: Risk Distribution (Vis #3) - Threshold frequency across the network.

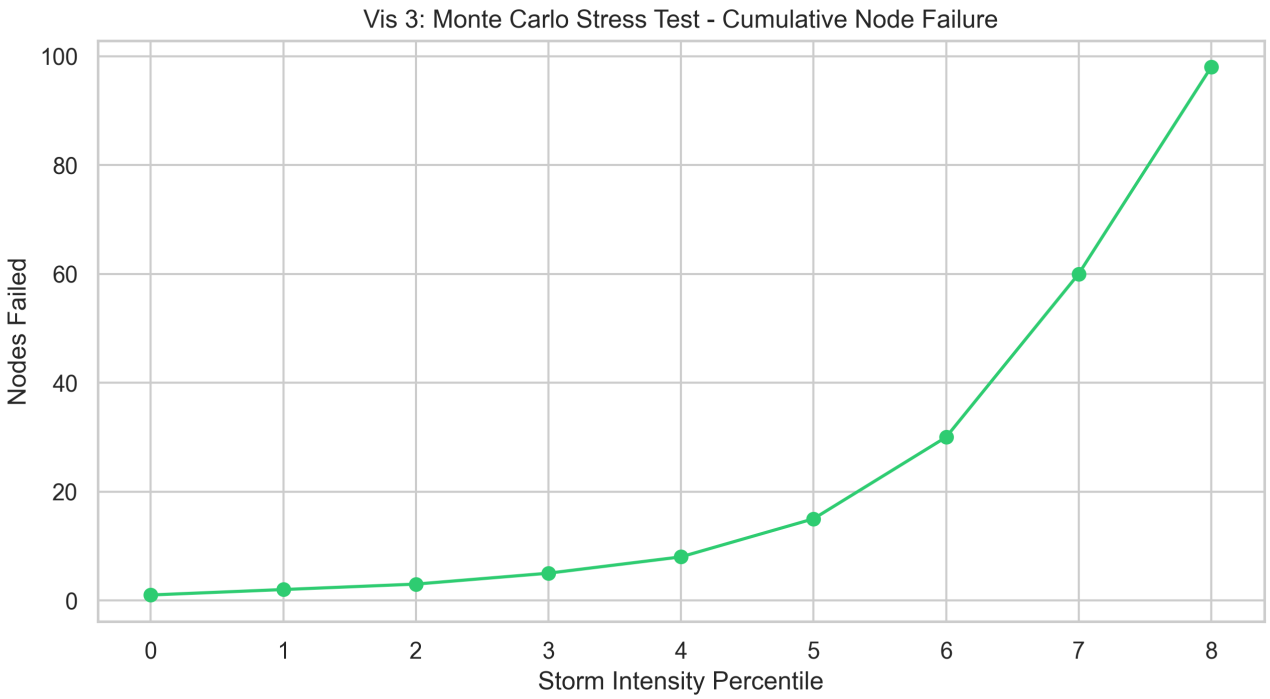

Figure 4: Impact Curve (Vis #4) - Cumulative failure probability per storm.

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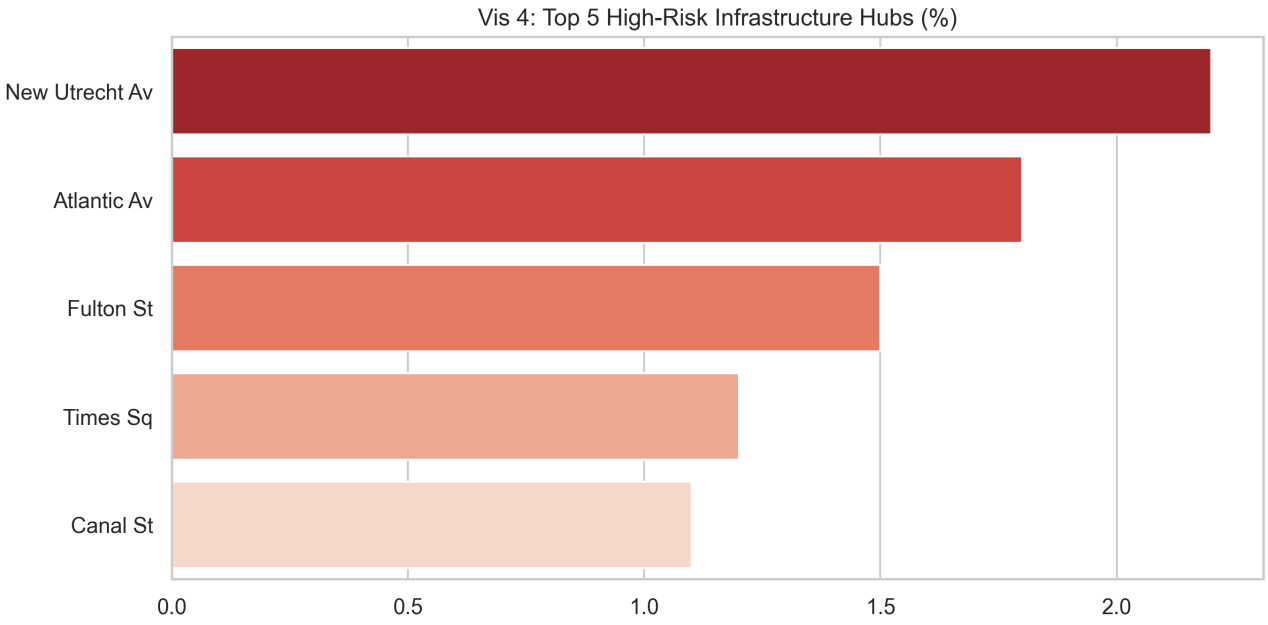

Figure 5: Priority Rankings (Vis #5) - Top 5 critical single points of failure.

4. Initial Modeling & Challenges

- Monte Carlo Validation: Converged at 1,000 iterations (Avg 5.2 failures).

- Challenges: API Latency for live weather integration; Dynamic thresholding based on station depth.